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Transcription regulation by biomolecular condensates

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Abstract

Biomolecular condensates regulate transcription by dynamically compartmentalizing the transcription machinery. Classic models of transcription regulation focus on the recruitment and regulation of RNA polymerase II by the formation of complexes at the 1–10 nm length scale, which are driven by structured and stoichiometric interactions. These complexes are further organized into condensates at the 100–1,000 nm length scale, which are driven by dynamic multivalent interactions often involving domain–ligand pairs or intrinsically disordered regions. Regulation through condensate-mediated organization does not supersede the processes occurring at the 1–10 nm scale, but it provides regulatory mechanisms for promoting or preventing these processes in the crowded nuclear environment. Regulation of transcription by transcriptional condensates is involved in cell state transitions during animal and plant development, cell signalling and cellular responses to the environment. These condensate-mediated processes are dysregulated in developmental disorders, cancer and neurodegeneration. In this Review, we discuss the principles underlying the regulation of transcriptional condensates, their roles in physiology and their dysregulation in human diseases.

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Introduction

The selective transcription of mRNA is a fundamental biological process that is often dysregulated in disease[1,2]. Decades of biochemical, structural and genetic studies have revealed the key proteins and nucleic acids involved in transcription regulation[3,4]. Classic models of transcription regulation focus on how these key proteins and nucleic acids form macromolecular complexes at the 1–10 nm length scale[5,6] (Fig. 1a). Imaging-based investigation of transcription in cells has revealed that these macromolecular complexes are further organized into compartments at the 100–1,000 nm length scale[7–20] (Fig. 1b). Several terms have been used to describe this larger-scale organization, including nuclear hubs, clusters, bodies, factories, microenvironments, foci and condensates, which, in principle, all refer to different facets of the same type of organization. Here, we will use the term biomolecular condensates, or more simply condensates, to (1) establish a simple dichotomy between molecular organization at the 1–10 nm scale (complexes) and at the 100–1,000 nm scale (condensates) (Fig. 1) and (2) emphasize the physicochemical principles borrowed from polymer `phase separation` that have guided the investigation of biomolecular organization at this larger scale.

Phase separation of proteins and nucleic acids had been known for decades[21–25], but the combined demonstration that cells utilize phase separation for organization of biomolecules[26] and to regulate biochemistry[27] ignited the current excitement about the role of phase separation in biology[28,29]. Foundational studies have focused on applying theory directly from polymer physics derived from simple systems[30–32]. In those simple systems, polymer phase separation occurs at `critical thresholds` of concentration, `multivalency`, strength of interactions, temperature or any other physicochemical parameters that influence the self-association of the polymer. Early studies of biological phase separation have revealed how these same parameters (multivalency, strength of interactions, temperature and so on) regulate the formation of biomolecular condensates of proteins and nucleic acids in purified reconstitutions[27,33–39] and in cells[27,39–42]. Although the heterogenous and non-equilibrium environment of the cell and the complex composition of nuclear condensates[7,11,43] cause deviations from theories derived from simple experimental systems[44], integrating biologically relevant parameters into theory continues to enable new hypotheses, models and experiments. In this way, principles borrowed from polymer phase separation continue to support progress towards deciphering how condensate-mediated organization regulates cellular biochemistry[45,46]. The focus of this Review is on condensate-mediated regulation of RNA polymerase II (Pol II) transcription.

The organization of the Pol II regulation machinery (including transcription factors, cofactors, Pol II and dozens of Pol II-associated proteins) into condensates, which for conciseness we will refer to as '`transcriptional condensates`', is driven by dynamic multivalent interactions among DNA, protein and RNA. Although there are many potential sources of multivalent interactions, the best-studied ones are multivalent `domain–ligand pairs`[27,47–50] and intrinsically disordered regions (IDRs) of proteins[33,34,36,38]. Multivalent surfaces, which are provided by `DNA regulatory elements` such as promoters and enhancers together with associated chromatin and RNA, are key regulators of condensate

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formation at specified locations[40,48,51–53]. Concepts borrowed from phase separation provide a framework to understand and investigate how dynamic multivalent interactions among components of the transcription machinery organize and regulate transcription in cells. As we discuss in the next section, current experimental data support a model in which the complexes (1–10 nm scale) required for the multi-factor and multistep process of regulating Pol II are organized into condensates (100–1,000 nm scale) that promote or prevent transcription of specific genes within the crowded environment of the nucleus[7,8,11,19] (Fig. 2).

In this Review, we discuss recent evidence linking condensate-mediated organization to transcription regulation, describe the mechanisms of transcriptional condensate formation, highlight how these mechanisms provide unique opportunities for regulating transcription, and examine how cells utilize these mechanisms to regulate transcription and how these mechanisms are disrupted in human disease. We conclude with discussing important challenges and questions that remain to be solved.

Condensates regulate gene activation

Precise control over which genes are expressed, at what levels and in what context requires hundreds of different molecules to operate at the various stages of the process of gene expression from DNA through RNA to protein. Condensate-mediated organization has been implicated in many different aspects of gene regulation[54–60]. In this Review, we mainly focus on the regulation of ‘gene activation’ by Pol II, which is defined as the steps of initiation through the assembly of the elongation-competent Pol II complex (Fig. 2a). Gene activation is regulated at several key steps and requires the concerted action of dozens of proteins[3,4] (Box 1). As described briefly in Box 1 and reviewed in detail by others[1–4], classic models of transcription regulation focus on how proteins assemble to form complexes that carry out each step of the process (Fig. 2a). We now have structural models of many of these key steps, which reveal at atomic resolution the mechanisms of regulation in cell-free reconstitutions[5,6,61–63]. By contrast, a condensate model of transcription regulation focusses on how the various complexes are spatially organized in the crowded nucleus[7,8,11,64,65] (Fig. 2b,c). As we discuss below, current data support a role for condensate-mediated organization in regulating gene activation in cells. It must be noted that the assembly of any individual complex, and any step in the process, is possible without higher-level organization, but the question is whether the entire process of gene activation can occur at a meaningful rate inside cells without higher-level organization. Although it remains unclear whether all transcriptional events in cells require condensate-mediated organization, for the many genes so far investigated in detail, condensate-mediated organization has been shown to regulate transcription by promoting (Fig. 2b) or preventing (Fig. 2c) the multistep process of gene activation.

Experiments in diverse systems ranging from in vitro reconstitution to live-cell imaging in developing organisms have begun to directly link organization by condensates to the regulation of transcription. Larger-scale organization of transcription is a long-standing observation, with early studies in fixed cells giving the impression that stable ‘factories’ of transcription form in the nucleus, to which genes translocate[66–68]. Super-resolution

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live-cell imaging methods found that these clusters of Pol II and other components of the transcription machinery (Table 1) are highly dynamic[69–72]. Early evidence that the dynamic clustering of Pol II regulates transcription came from live-cell imaging of Pol II clusters at the β-actin locus in serum-stimulated cells[73], wherein the number of Pol II molecules in the cluster correlates with mRNA output. This observation was supported by further studies of higher-order clustering of Pol II and multiple cofactors[74]. In line with these findings, several in vitro reconstitution experiments have demonstrated a direct link between condensate-mediated organization and Pol II transcription[75–77]. In cells, synthetically induced condensates amplify transcription[78] or directly activate transcription when targeted to specific genomic loci by different methods[79,80]. Reporter assays to visualize developmentally regulated gene transcription in live *Drosophila melanogaster* embryos demonstrated that condensate formation is coincident with or immediately precedes transcription and that manipulating this larger-scale organization affects transcription[81]. Live-cell imaging in mouse embryonic stem cells demonstrated that the proximity of condensates to the endogenous *Sox2* gene regulates its transcription[82]. The formation of condensates by the cell-type-specific transcription coactivator myocardin, is directly linked to its ability to activate cell-identity genes[83]. These key examples ranging from in vitro reconstitutions to live-cell imaging highlight the growing body of evidence that condensate-mediated organization of the transcription machinery can regulate transcription in diverse experimental systems and biological contexts.

The functional connection between condensate-mediated organization and gene activation is now well established (Table 1). Major open questions relate to the precise mechanisms by which condensates carry out these functions and the extent to which these mechanisms are used by cells and organisms to regulate the transcription of specific genes. The best-studied mechanism of condensate function in transcription is the selective compartmentalization (also known as selective partitioning) of the regulatory machinery at specific locations within the cell, either to concentrate them at target genes leading to activation (Fig. 2b) or to sequester them away from target genes leading to repression (Fig. 2c). Selective compartmentalization by condensates creates local environments that can, for example, exclude negative regulators and include positive regulators of transcription, thereby promoting higher rates of gene activation[84]. Conversely, certain condensates may reduce gene activation rates by sequestering positive regulators away from genes[85] or attracting corepressors to the regulatory elements of genes[86]. In the next section, we discuss how recent studies of the mechanisms underlying condensate formation and selective compartmentalization are beginning to reveal the regulatory potential of condensates.

Formation and composition of transcriptional condensates

From the outset of molecular descriptions of gene activation, research has focused on how individual molecules specifically bind to DNA to either direct or prevent RNA polymerase binding. The characterization of how and where these complexes form and how this process is regulated by chromatin has been the main driving pursuit for the field. By contrast, a condensate perspective is focused on the larger-scale organization that creates local concentrations of these complexes to promote or prevent transcription. The simplest model of how transcriptional condensate formation and function are regulated must account for (1)

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the specificity of genomic locus and (2) the specificity of condensate composition. In the context of this model, where a condensate forms — for example, on or off chromatin, and if on chromatin, then at what locus — and the complement of proteins or other biomolecules that are partitioned by the condensate are both important for understanding the function of a transcriptional condensate. These two layers of condensate regulation use both classic structured interactions and dynamic multivalent interactions to create distinct localized environments, starting with structured domains that bind to DNA or chromatin and building up into multivalent networks of interactions that locally concentrate the components of the Pol II regulatory machinery (Fig. 3a). Dynamic regulation is provided by the accessibility of regulatory elements and by reversible post-translational modifications (PTMs) of histones[87,88], DNA-binding transcription factors (TFs)[87], Pol II[15,16] and cofactors[89] (Table 1). Whereas the important role of specific structured interactions in regulating transcription and their underlying mechanism of action are well-established and reviewed in detail by others[1–6], the dynamic multivalent interactions responsible for condensate-mediated transcription regulation is an emerging field of research and is, therefore, the main focus in this Review. In this section, we discuss the principles of condensate formation and their composition specificities with a focus on the role of dynamic multivalent interactions, and the unique regulatory opportunities provided by condensate-mediated regulation.

Locus specificity

Transcriptional condensates form at specific genomic loci by a dynamic network of interactions seeded by the ability of TFs to simultaneously bind DNA, RNA and protein[90]. DNA regulatory elements contain clusters of binding sites that provide a multivalent template for TFs to bind[51,52]. TFs expand on this multivalency by engaging in TF–TF and TF–cofactor interactions, which are further expanded by cofactor–cofactor and cofactor–chromatin interactions (Table 1 and Fig. 3a). As TFs do not bind directly to Pol II or to many of the factors required for gene activation (for example, factors required for promoter escape, pause release, and elongation), TF binding to DNA is the first and often insufficient step towards gene activation, which ultimately requires a network of coactivator-mediated interactions to locally concentrate the many components of the transcription machinery. In this context, the main role of TFs is to ‘translate’ the language of DNA regulatory elements into the language of protein–protein interactions. In addition to TFs, chromatin at active regulatory elements is marked by a specific set of histone PTMs[87,88] and by the transcription of nascent RNAs[77,91–93], both of which can act as multivalent seeds for condensate formation[77,91,94] (Fig. 3b).

The combination of multivalent interactions and structured interactions at clusters of binding sites on DNA enables condensate formation at specific loci. This process of localized condensate formation has been described as seeded nucleation[40,43] or as pre-wetting transition[52] and is an important aspect of locus specificity. This feature of transcriptional condensates ensures that condensates only form at the seeded locus when the correct set of transcription factors is expressed in that cell and bound to the locus. This reliance on a heterotypic network of interactions wherein an obligate member of the network is a DNA regulatory element or cluster of regulatory elements is proposed to enable localized and selective condensate formation. Such formation of transcriptional condensates was

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experimentally demonstrated by careful manipulations of nuclear concentrations of the transcription cofactor ENL. This work has shown that at endogenous concentrations, ENL formed chromatin-bound condensates, but when the concentration was too high, homotypic interactions dominated and ENL formed condensates with itself off chromatin, which inhibited transcription[95]. Overexpression of other factors can also lead to non-functional condensates forming off of chromatin[96,97]. These findings suggest that a finely tuned balance of interactions is required for the formation of functional condensates and highlight the importance of studying these processes at endogenous concentrations of condensate components. Given this dependence on regulatory elements, the size of transcriptional condensates may be determined, in part, by the length of the underlying DNA regulatory element sequence. Evidence in support of this idea was shown for the `histone locus body`, wherein manipulating the number of regulatory elements that seed the histone locus body directly affected the size of the condensate formed[98]. Beyond the regulation of size, whether or not a condensate is formed at a locus can be determined, in part, by the number of TF-binding sites present in a regulatory element, with condensate formation being favoured at critical thresholds of number and density of TF-binding sites[51]. The presence of multiple binding sites can considerably reduce the concentrations required for certain factors to undergo phase separation. For example, nucleosome arrays and Polycomb repressive complex 1 (PRC1) work together to enhance PRC1 condensation, lowering the necessary concentration for this process to about 1/20 of that required by PRC1 alone[59]. Together, these findings suggest that the location of condensate formation can be regulated by both the concentration of specific DNA-binding or chromatin-binding proteins and the architecture of regulatory DNA.

Composition specificity

Transcriptional condensates selectively partition biomolecules through an ensemble of multivalent interactions. Instead of investigating how a single molecule interacts with another molecule, a condensate model focuses on how multiple interacting molecules form a network that selectively partitions other molecules. The complement of biomolecules recruited by TFs and local chromatin creates a network of multivalent interactions that can selectively compartmentalize factors. In the past, it is probable that these functional but context-dependent interactions were largely missed owing to the use of standard biochemical purification strategies, which often rely on high degrees of dilution and high-stringency washes, which skew discovery towards stable structured interactions operating at the 1–10 nm scale. Use of recently developed methods that either investigate interactions in situ or reconstitute condensates in vitro has begun to uncover these overlooked yet functional interactions that enable the formation of compositionally distinct condensates[84,99–101] (Fig. 3c). Two major types of protein features have been extensively studied in this context: IDRs and multivalent domain–ligand pairs (Fig. 3d).

Dynamic multivalent interactions involving IDRs have often been described as nonspecific[102], but emerging evidence demonstrates that, in the context of condensates, interactions between IDRs can be specific. Candidate-based methods testing pair-wise interactions of IDRs of FET (FUS, EWS, TAF15) protein[99,103,104] or of TFs and coactivators[65,105,106] revealed that there is specificity encoded into IDR sequence features.

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Unbiased proteomics analysis of transcriptional condensates combined with experiments in vitro and in cells have revealed that interactions between different IDRs were both necessary and sufficient for selective partitioning of positive regulators of transcription by the IDR of MED1, a subunit of the `Mediator coactivator complex`, and for gene activation[84]. These particular IDR–IDR interactions are driven by multiple alternating `patterned charge blocks` in their amino acid sequence. These patterned charge blocks in IDRs are sufficient to retarget complexes and change gene activation in cells. IDR swapping experiments with heterologous IDRs of similar or divergent sequence features have provided evidence that the specificity encoded in IDRs matters for function[84]. These charge blocks are just one type of specificity encoded by IDRs. Other sequence features encoded by IDRs use different enrichments and patterns of amino acids to encode various composition specificity rules, which are still being uncovered[101]. Interestingly, swapping between condensate-forming IDRs that do not have similar sequence features can rescue condensate formation but not function, probably owing to the inability to selectively partition the right set of proteins[83,100]. IDRs have also been shown to be sufficient for targeting proteins to specific genomic loci independently of structured domains[106–109], suggesting that IDRs are being targeted to compositionally distinct condensates nucleated at these loci. Although rules have begun to emerge for how some IDRs encode specificity[84,100,101,106,110], there is probably more left to uncover.

Although the literature has been dominated by studies of the role of IDRs in condensate formation and composition, the earliest work on selective partitioning by condensates has focused on multivalent domains[47,48]. Both IDRs and multivalent domains form condensates by the same principles of dynamic multivalent interactions[111], but whereas for IDRs the interactions are presumed to occur between individual amino acids or regions of clustered amino acids[106], for multivalent domains the interactions are between a structured domain and a ligand. Many cofactors and cofactor complexes function predominantly as hubs of multivalent interactions owing to their many interaction domains[112]. These multivalent interactions include domains responsible for interacting with specific histone PTMs (PHD, bromo, YEATS, chromo and so on), with the activation domains of TFs (KIX, TAZ, IBiD, RID and so on), and with dimerization or polymerization domains. These interactions typically involve a structured domain of a transcription cofactor with a motif in an IDR of TFs (activation domains), histones (tails), Pol II (carboxy-terminal domain) or another cofactor. Several studies have shown that multivalent interactions with histone PTMs are important for condensate formation[54,113–118]. Recent single-particle tracking experiments of the coactivator p300, which is a coactivator and histone acetyltransferase, have demonstrated that its chromatin occupancy was determined by multivalent interactions of its TF-interacting domains[119].

Beyond these classic interaction domains, new classes of domain–ligand pairs have recently been described for proteins involved in transcription elongation, in which structured domains interact with specific motifs within IDRs[120]. It is probable that other domain–ligand pairs remain to be identified and that these multivalent and context-dependent interactions have been missed by classic biochemical studies. Intriguingly, many of these domain–ligand interactions are regulated by PTMs. Additionally, different types of multivalent

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interactions synergistically influence the properties of condensates. For instance, the critical concentration for PRC1 condensate formation is determined by the interaction between nucleosome arrays and PRC1, heterotypic interactions between different PRC1 subunits, and polymerization of PHC2 (which mediates PRC1 clustering)[59,117,121,122] (Table 1). Meanwhile, the morphology, dynamics and stability of PRC1 condensates are determined by the polymerization of PHC2 (ref. 59). Although multivalent domain–ligand pairs in transcriptional condensates are not as well studied as IDRs, they probably have a substantial role in the organization of the transcription machinery by condensates.

Regulatory possibilities unique to condensates

The hyper-cooperative nature of condensate formation leads to regulation through thresholds of concentration of TFs or cofactors, accessibility of regulatory elements, and changes to multivalency owing to ligand-binding or PTMs. This complexity offers novel regulatory mechanisms that drive the system above or below these thresholds (Fig. 3e). The classic models of transcription focus on the cooperative assembly of subunits into complexes, and in particular TF–DNA complexes. The condensate model focusses on the cooperative assembly of complexes into condensates, which enables the rapid and dynamic organization of an entire biochemical process instead of the cooperative assembly of a single complex. As discussed in the ‘Locus specificity’ section above, TF–DNA interactions are probably required for condensates formation, but they are only a first step in gene activation. Similar to the parameters that determine whether polymers undergo phase separation, condensate formation in cells is regulated by the degree of multivalency, the strength of interactions, concentration, temperature and any other feature that modulates the network of interactions. These parameters are all biologically tuneable and have been shown to regulate condensate formation and in turn transcription (Fig. 3e). The number of transcription-factor binding sites above thresholds nucleates condensates and activates transcription reporters in cells[51] (Fig. 3f). Myocardin, a cell-type-specific transcription coactivator, is able to activate new gene programmes only when its nuclear concentrations is above the concentration threshold for condensate formation[83] (Fig. 3e). Multiple nuclear hormone receptors exhibit enhanced multivalency upon hormone (ligand) binding, resulting in nucleation of their condensates and activation of target gene transcription in cells[123–125]. In plants, temperature regulates condensate formation with functional consequences on transcription[126].

There are also non-intuitive consequences of heterotypic networks, in which optimal concentrations favour condensate formation, but increasing the concentration of a single component of the network disfavours condensate formation (Fig. 3g). This is referred to as a ‘`re-entrant phase transition`’ and is best understood for oppositely charged polymers that include positively charged proteins and negatively charged RNA[127]. These re-entrant phenomena are proposed to regulate the formation of transcriptional condensates by promoting condensate formation at low rates of RNA synthesis up to a point of charge imbalance, beyond which higher rates of RNA synthesis disfavour condensate formation[77]. This process provides an elegant mechanism for negative feedback control[46].

These types of regulation, which are unique to the condensate model, are used to selectively regulate transcription in physiological contexts and are dysregulated in disease. In the

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next sections, we discuss how condensate-mediated organization of the transcriptional machinery is used in the regulation of developmental gene programmes and in response to environmental signals, and how this process malfunctions in human disease.

Regulation of development by transcriptional condensates

A fundamental aspect of development is precisely timing the transcription of key developmental genes, a process crucial for cell-fate determination. Recent studies have shown that transcriptional condensates offer a potential mechanism for this process of decision making by regulating transcription above or below thresholds of condensate formation (Table 1). This coupling of gene activation to thresholds of condensate formation, together with other concurrent processes operating at the 1–10 nm scale (transcription-factor-binding cooperativity, chromatin accessibility and so on), may provide a 'switch-like' response to changes in nuclear concentrations of key regulators or other physiologically tuneable variables (Fig. 3e). In this section, we discuss how transcriptional condensates regulate development in animals and plants.

Animal development

The cell-type-specific transcription coactivator myocardin forms condensates in cardiomyocyte and in smooth muscle cells at sites of cell identity gene transcription[83]. During differentiation, myocardin forms condensates at threshold concentrations providing a switch for activation of its own gene and of cell-type-specific gene programmes, guided by the DNA-binding transcription factor serum response factor (SRF). In reporter and reprogramming assays, the ability of myocardin to activate genes was shown to depend on its ability to form condensates. The proposed mechanism of condensate activity was the selective partitioning of p300 and Pol II by a C-terminal IDR of myocardin[83]. The formation of condensates at critical concentrations, the condensate-dependent transcriptional activity, and the ability of myocardin to activate its own gene constitute a condensate-mediated feedforward loop driving cell-fate determination (Fig. 4a). Given that many cell type regulators also positively regulate expression of their own gene and are highly concentration dependent[128], it is probable that other cell fate decisions are made using similar concentration-triggered condensates.

Condensate formation has been implicated in cell state transitions in several other contexts. For example, SS18 (also known as SSXT) condensates have a role in the differentiation of pluripotent cells into somatic cells[129], and CCAAT/enhancer-binding protein-α (C/EBPα) condensates may aid in the *trans*-differentiation of B cells to macrophages[130] (Table 1). Beyond recruitment of the transcription machinery, condensates formed by pioneer TFs can alter chromatin structure. Notably, the pioneer factor FOXA1, through its capacity for DNA binding and for establishing multivalent interactions with its IDRs, can form condensates to open condensed chromatin[131–133]. This process, which is accompanied by changes in histone modifications, increases the levels of acetylated histone H3 Lys27 (H3K27ac) and monomethylated histone H3 Lys4 (H3K4me1) while decreasing H3K9me3 level, thereby activating genes essential for cell-fate determination[133] (Fig. 4b). However, whether the chromatin opening caused by FOXA1 condensates directly leads to these

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histone modifications, or whether FOXA1 and specific modifying enzymes act together to open the chromatin, remains unclear. Similarly, EBF1 (also known as COE1), another pioneer factor, forms condensates that recruit components of the BAF complex and cluster EBF1 target genes, enabling BAF-mediated chromatin opening, gene expression and B lymphopoiesis[134]. In embryonic stem cells, CTCF–RYBP condensates are associated with enhanced self-renewal capacity and a reduced tendency to differentiate towards neural progenitor cells[135]. Transcriptional condensates containing pluripotency and self-renewal transcription factors such as OCT4 (ref. 136), KLF4 (ref. 137) or NANOG[138] are implicated in cell-type reprogramming processes (Table 1). Furthermore, a recent study has identified specific transcription factors with dual roles as activators and repressors, which are pivotal in regulating developmental processes[139]. These dual-role TFs form nuclear condensates that selectively partition core transcription units, thereby maintaining low-level expression of genes involved in development that are typically highly activated; by contrast, they enhance the expression of genes involved in stem cell maintenance, which are originally expressed at low levels[139]. This dual role ensures gene expression at intermediate levels, which is essential for maintaining stable states in stem cells (Fig. 4c). Another study, focused on investigating the molecular determinants of TF condensate formation, has found that the periodic spacing of aromatics within many TF IDRs was suboptimal for condensate formation. Rearrangement of the aromatics to have optimal spacing improved the capacity of TFs to form condensates and enhanced the activity of these TFs in both luciferase reporter assays and in macrophage reprogramming (C/EBPα) and neural differentiation (NGN2)[140]. The suboptimal condensate formation programmed into TFs highlights again how finely tuned the network of interactions underlying condensate formation is to regulate when and where condensates form in the regulation of developmental gene programmes.

In vivo evidence further supports the role of transcriptional condensates in the regulation of animal development[141]. Research using quantitative single-cell live-imaging and computational analysis has revealed that the dynamics of transcriptional condensate formation and disassembly are intricately linked to `transcription bursting` in developing *D. melanogaster* embryos[81,142]. These condensates have been identified as coactivators of distant genes and their formation is mediated by multivalent protein–protein interactions between TF IDRs[81]. Intriguingly, the artificial introduction of a `poly-glutamine` (polyQ) tract to TFs, such as Bicoid, promotes the formation of condensates by facilitating interactions through multiple hydrogen bonds. These additional, ectopic TF condensates enhance transcription bursting by promoting more frequent and intense interactions at gene loci, which in turn promotes the expression of the genes *hunchback* and *knirps*, leading to segmentation defects during embryogenesis[81]. Similarly, genetic perturbations and real-time imagining discovered that the pioneer TF Zelda initially forms small and unstable condensates in *D. melanogaster* embryos[143,144]. These condensates of Zelda collaborate with the transcription coactivator Cbp to support the formation of Brd4 condensates, which show little colocalization with Zelda, but aid in the formation of Pol II condensates, thereby initiating transcription at target regions[144] (Table 1). Similar phenomena have been observed in early zebrafish embryogenesis wherein the TF Nanog forms condensates driven by its IDRs, subsequently recruiting Sox19b and Pol II in sequence, thereby promoting the expression of genes essential for development[145].

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Together, cells utilize condensate-mediated organization of the gene control machinery to establish gene regulation networks that are both robust and able to decisively change in response to crucial cues. The activation or inactivation of an entire biochemical process at thresholds afforded by condensate-mediated organization provide a potential mechanism for the switch-like and decisive changes in gene expression observed during animal development.

Plant development

The development of plants, including germination, seedling growth, flowering and seed production, is intricately governed by gene expression in response to external cues such as temperature, light and hormones. Plant development is primarily regulated through the modulation of transcription, either by promoting the formation or dissolution of transcriptional condensates, or by selectively sequestering or recruiting key regulators. Interestingly, for several of the examples investigated so far, and in contrast to the examples studied in animal development, plants utilize condensate formation off chromatin to regulate transcription (Table 1).

Transcriptional changes in response to temperature fluctuations are a crucial aspect of plant physiology. Two key examples highlight the role of condensates in temperature sensing. EARLY FLOWERING 3 (ELF3) is a plant growth regulator and transcription repressor that is organized into condensates at temperatures above 27 °C (ref. 126). This process regulates flowering timing by enabling ELF3 to sense and respond to temperature changes. At 22 °C, *Arabidopsis thaliana* ELF3 is dispersed in the cell, binds to DNA and inhibits the transcription of developmental genes. At 27 °C, ELF3 forms condensates off chromatin, which prevents gene suppression and results in plant growth[126,146,147] (Fig. 4d). The transition of ELF3 from a dispersed state to forming condensates is characteristic of `lower critical solution temperature behaviour`[148,149]. *A. thaliana* FRIGIDA, a transcription activator and key regulator of vernalization[150], acts as a sensor of low temperatures. Cold temperatures increase FRIGIDA concentration through changes to protein stability, resulting in the formation of condensates that sequester FRIGIDA away from the gene encoding FLOWERING LOCUS C (FLC), a major floral repressor, thereby reducing its expression. Although this mechanism does not directly induce flowering, it primes the plant for flowering when conditions become favourable[151,152] (Table 1).

Light regulates plant development through a complex network of photoreceptors that are sensitive to various wavelengths. Certain photoreceptors form condensates referred to as photobodies, which attract transcription-related proteins to inhibit or activate gene transcription. For instance, red light triggers a conformational change in phytochrome B (phyB) that results in its translocation and accumulation in the nucleus, leading to the assembly of photobodies through multivalent interactions of phyB[153]. Photobodies attract and either sequester or degrade TFs — phytochrome-interacting factors — which regulate the development of seedlings in darkness. These mechanisms are used by transcriptional condensates in plants to regulate development in response to light[153–160] (Fig. 4e). Similarly, blue light activates the photoreceptor cryptochrome 2 (CRY2) to form photobodies. These CRY2-associated photobodies recruit TFs such as TCP22, which drive the expression

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of genes essential for plant growth, development and regulation of circadian rhythms[161–164]. This intricate interplay between light, photoreceptors and the transcription machinery highlights the role of condensate-mediated organization in plant adaptation to environmental conditions (Table 1).

Plant growth and development are also tightly controlled by hormones such as auxin. Auxin binding to specific receptors initiates the degradation of auxin-responsive IAA proteins, thereby freeing auxin response factors (ARFs) to regulate genes that are crucial for growth and development. Intriguingly, recent studies indicate that in plant regions with reduced auxin sensitivity, such as the mature upper root, cytoplasmic condensates sequester ARFs and thus restrain the transcription of auxin-responsive genes[85,165–167] (Fig. 4f).

Responses to environmental stimuli that are mediated by transcriptional condensates

Cells continuously monitor and adapt to environmental stimuli such as stresses and intercellular signalling, by changing their transcriptional profiles. Recent research has underscored the role of transcriptional condensates in the selective regulation of target genes in response to such stimuli. The formation of these condensates is triggered by environmentally responsive changes in the concentration or multivalency of key proteins (Table 1). It is crucial to distinguish that, unlike the relative long-term influences of environmental cues on development discussed in the previous section, the environmental stimuli we discuss here provoke acute and reversible cellular responses.

Environmental stress

Heat-stressed cells upregulate the expression of heat shock proteins (HSPs) while downregulating global gene expression. In normal conditions, heat shock factor 1 (HSF1) remains inactive in the cytoplasm, sequestered by HSPs. Following heat stress, misfolded proteins sequester HSPs away from HSF1 (ref. 168). HSF1 translocates into the nucleus and forms condensates driven by multivalent interactions between its IDR, the Mediator coactivator complex and Pol II[169]. Through HSF1 binding to multiple heat shock elements at target genes, HSF1 condensates enhance the expression of *HSP* genes, thereby orchestrating a cellular response that mitigates the damaging effects of heat[168,169] (Fig. 5a). Simultaneously, HSF1 condensates enhance the transcription of satellite III long non-coding RNA, which helps to suppress global gene expression by sequestering crucial transcription factors into nuclear stress bodies, thereby providing additional cell protection[170]. Upon stress relief, these condensates are dissolved by HSPs[171]. By contrast, to globally reduce transcription during heat stress, the transcription regulator negative elongation factor (NELF) forms condensates facilitated by multivalent interactions, including between its SUMO interaction motifs and heat-stress-induced SUMOylated proteins, such as NELF itself and ZNF451, as well as interactions with its IDR. These condensates then interact with Pol II and repel P-TEFb and other elongation factors, ultimately leading to the downregulation of target genes[89,172] (Fig. 5b).

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Transcriptional condensates are also crucial for osmotic stress response. Cleavage and polyadenylation specificity factor 6 (CPSF6) is required for mRNA maturation by facilitating the cleavage of its 3′-end and the addition of a poly(A) tail. Reduced cell volume in hyperosmotic conditions leads to an increase in CPSF6 concentration, which triggers CPSF6 phase separation, potentially driven by multimerization[173,174] or by multivalent interactions with its C-terminal arginine/serine-like domain[175]. CPSF6 condensates partition the pre-mRNA cleavage complex away from transcription end sites, leading to transcription termination defects and abnormally elongated transcripts[173,174] (Fig. 5c). In plants, the transcription regulator SEUSS responds to osmotic stress by altering the conformation of its IDR1 domain, potentially facilitating intermolecular interactions that promote condensate formation to regulate stress-responsive genes[176] (Fig. 5d).

Oxidative stress, although typically harmful, can sometimes serve as a regulatory mechanism of normal physiological processes. Hydrogen peroxide (H_2O_2) induces condensate formation of the transcription factor TERMINATING FLOWER (TMF) through multivalent interactions including strong covalent bonds between intermolecular Cys residues, weak interactions between the IDRs, and electrostatic interactions between TMF and DNA. These condensates are located at the *ANANTHA* gene promoter, suppressing *ANANTHA* expression and ultimately influencing the development of floral organs[177,178] (Fig. 5e).

Intercellular signalling

Cells respond also to external signalling molecules through specific protein condensates that alter the expression of relevant genes. For instance, guanylate-binding protein-like GTPase 1 (GBPL1), characterized as a pseudo-GTPase, binds tightly to heterochromatin and constrains the accessibility of GBPL3 to chromatin, thus limiting its transcriptional activation of immunity genes[155,179]. Following exposure to immune system signals, such as salicylic acid and pipecolic acid, or to pathogens, cytoplasmic GBPL3 is activated and translocates into the nucleus, thereby increasing its nuclear concentration to form condensates through multivalent interactions of its IDR, assisted by its GTPase activity. These condensates gather at defence gene promoters to upregulate transcription by recruiting Pol II and Mediator subunits[179] (Fig. 5f).

Nuclear receptors are a unique class of TFs that are activated by ligands and can form condensates to regulate transcription. When bound to glucocorticoids, the glucocorticoid receptor undergoes a conformational change and enters the nucleus. The high local concentration of glucocorticoid receptor at glucocorticoid response elements induces condensate formation, predominantly driven by multivalent interactions between its IDRs. Intriguingly, the specific glucocorticoid response elements wherein these condensates are located influence the selective recruitment or exclusion of specific coregulators, ultimately dictating the upregulation or downregulation of target genes[125,180–182]. Similarly, several other nuclear receptors form condensates that regulate transcription, including androgen receptor[124,183], oestrogen receptor α[92,105,184] (Table 1) and peroxisome proliferator-activated receptor γ, which dimerizes with retinoid X receptor α[185]. However, whether these condensates are merely byproducts of receptor activation by ligands or essential for

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transcription activation, and whether they affect all their target genes or just a subset of them, is still unclear.

Certain cytokines can induce transcription cofactors to form functional condensates. For instance, in normal conditions, without interferon-γ (IFNγ) signalling, histone acetyltransferase KAT8 localizes to the promoter region of *PDL1* and acetylates H4K16, thereby activating *PDL1* expression at a basal level[87]. In the tumour microenvironment, elevated levels of IFNγ activate transcription of the interferon regulatory factor 1 (*IRF1*) gene. IRF1 in high concentration forms condensates with KAT8, driven by the structured-domain interactions between the DNA binding domain of IRF1 and KAT8, and by multivalent interactions between their IDRs[87]. These KAT8–IRF1 condensates enhance acetylation of both H4K16 (thereby increasing chromatin accessibility) and IRF1, resulting in high-affinity interactions with the *PDL1* promoter and recruitment of transcription-associated factors for transcription upregulation[87] (Fig. 5g). Additionally, signalling pathways such as transforming growth factor β (TGFβ) and tumour necrosis factor (TNF) can promote the formation of transcriptional condensates that facilitate the expression of target genes. For instance, in neural stem cells stimulated by TGFβ, activated SMAD3 enters the nucleus and forms condensates with JMJD3, promoting the activation of genes responsible for neurogenesis[161]. Similarly, in human SK-N-AS cells, TNF induces entry of the NF-κB subunit p65 to the nucleus and formation of co-condensates with the p300 and BRD4 to upregulate target genes[186] (Table 1).

Transcriptional condensates in disease

In human disease, the regulation of the transcription machinery by condensates is often disrupted[187,188]. Understanding how diseases are driven by condensate dysregulation could provide new mechanistic insights into the functional role of condensates. Ultimately these insights could suggest strategies for therapeutically targeting the underlying disease. Four broad mechanisms of condensate dysregulation in disease are emerging: (1) gain of new interaction networks, (2) disruption of interaction networks, (3) changes to nuclear concentration of proteins, and (4) disrupted signalling. In this section, we discuss key examples of each of these mechanisms (Table 1).

Gain of interaction networks

Condensate-mediated organization and regulation of transcription require a precise network of heterotypic interactions. Two broad classes of genetic lesions have been found to lead to gain-of-function through enhancing or creating new interaction networks: (1) translocations that fuse an IDR to a DNA-interacting or chromatin-interacting protein, leading to the expression of a chimeric oncoprotein and (2) small insertions or deletions (indels) in structured regions.

IDR oncofusion proteins. Chromosomal rearrangements that fuse two genes to encode for a novel fusion protein are observed in 17% of all cancers[189]. A major class of these oncoproteins brings together an IDR and a DNA-interacting or chromatin-interacting domain[190–192]; they can cause disease through several mechanisms, including the loss or gain of interaction networks[193]. Recent studies have found that some of

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these oncofusions enable multivalent interactions that drive condensate formation, often attracting many new components and potentially leading to pathological dysregulation of transcription. For example, FET fusion proteins, commonly associated with sarcomas and leukaemias, undergo phase separation and form condensates at target genes[75,194–198] (Fig. 6a). These condensates partition Pol II and transcription cofactors, thereby disrupting normal transcriptional processes, reactivating silenced genes and promoting tumorigenesis[196]. Similarly, SS18–SSX is a hallmark fusion protein of synovial sarcomas. The QPGY domain in the SS18 IDR promotes the formation of SS18–SSX condensates and the partitioning of the BAF complex, thereby enhancing gene expression and driving synovial sarcomagenesis[199]. Condensates formed by nucleoporin (NUP) IDR fusion proteins, such as NUP98–KDM5A, NUP98–LNP1, NUP98–PRRX1 and NUP98–NSD1, effectively partition transcription cofactors[200–203]. Additionally, NUP98–HOXA9 condensates facilitate chromatin looping independently of CTCF and cohesin, proteins typically assumed to regulate enhancer–promoter contact frequency, thereby creating large clusters of enhancers similar to super-enhancers, which can promote oncogene expression[204] (Table 1). BRD4–NUT, the driver of NUT midline carcinomas, binds to acetylated chromatin to form condensates. These condensates partition p300, which spreads acetylation over megadomains of chromatin and thus opens its 3D structure and promotes transcription[205–208] (Fig. 6b). In the examples of transcription-associated fusions studied so far, the enhanced partitioning of positive regulators of transcription is a common mechanism of gene activation, but exactly how these condensate-promoting IDRs selectively partition the transcription machinery remains unresolved.

Indels in structured regions. The insertion or deletion of several amino acids in a protein can alter its conformation and interactions, leading to changes in condensate formation and composition. ENL, a subunit of the super elongation complex encoded by the *MLLT1* gene, interacts with acetylated histones through its YEATS domain and is crucial for recruiting the complex to activated genes. Insertion and deletion mutations within the YEATS domain lead to structural alterations that enhance ENL's homotypic interactions and heterotypic interactions with other transcription cofactors, leading to upregulation of target genes, including the HOXA cluster, and contributing to the progression of Wilms tumour and acute myeloid leukaemia[95,209,210] (Fig. 6c). Although many oncofusions predicted to operate through formation gain-of-function condensates by IDR fusion have been catalogued[192], predicting which indel (or point mutation) will have similar effects as the indels of ENL is considerably more challenging. Whether additional examples of indels or other disease-associated mutations within structured regions of proteins operate through condensate-mediated organization such as ENL remains a compelling open question.

Loss of interaction networks

Genetic lesions that break the ability of a protein to engage in dynamic multivalent interactions or that disrupt the ability of other proteins to engage in these types of interactions have been implicated in cancer, developmental disorders and neurodegeneration.

Trinucleotide repeat expansions encoding Ala or Gln in disordered regions can disrupt the dynamics and composition of condensates. HOXD13 is essential for limb development in

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mammalian embryogenesis and expansions of an Ala repeat in HOXD13 are linked to synpolydactyly, a limb malformation[211]. Driven by the IDR, both normal and Ala expansion mutant HOXD13 form condensates; however, the mutant exhibits higher levels of homotypic interactions and lower levels of heterotypic interactions, evidenced by poor recruitment of transcription coactivators. This disruption to interaction networks, or 'condensate unblending', reduces HOXD13's transcriptional activity, resulting in lower expression of genes necessary for mesenchyme differentiation and digit formation[212] (Fig. 6d). Similar issues with condensates are seen in other disorders: Ala expansions in HOXA13 cause hand-foot-genital syndrome[212], Ala expansions in RUNX2 lead to cleidocranial dysplasia[212], polyQ expansions in TATA-box binding protein are associated with spinocerebellar ataxia type 17 (ref. 212), and polyQ expansions in androgen receptor result in prostate cancer and other androgen-receptor-involved diseases[213]. These developmental disorders highlight the importance and fine balance of dynamic multivalent interaction networks in establishing condensate composition and function to regulate transcription during development. Notably, the formation of abnormal condensates owing to identical amino acid repeat expansions is one of many mechanisms causing diseases. These expansions may also alter protein structure[214], impair its ability to interact with other proteins[215] and its localization[216], and lead to protein misfolding[216].

Nonsense mutations often result in truncated proteins with severely impaired functions. The Lys-specific demethylase (KDM) UTX (also known as KDM6A) is a tumour suppressor that can form condensates driven by multivalent interactions with its central IDR. These condensates actively demethylate H3K27me3 and simultaneously recruit various transcription cofactors, including the H3K4-monomethylating Lys methyltransferases (KMT) MLL3 (also known as KMT2C) or MLL4 (also known as KMT2D), and p300, which further modify histones and alter chromatin structure to regulate transcription, resulting in tumour suppression[217]. The UTX mutant lacking its central IDR does not form these transcriptional condensates and cannot effectively function as a tumour supressor[217] (Fig. 6e). Several IDR truncations are linked to developmental disorders, suggesting that the truncation of IDRs could have broader roles in disease biology.

Genetic lesions that create novel coding sequences through repeat expansion or frameshifts have been shown to disrupt the condensates formed by other proteins *in trans*. Expansion of the hexanucleotide repeat GGGGCC in the intron of *C9orf72* is a driver of amyotrophic lateral sclerosis. This repeat encodes a toxic di-peptide GR or PR repeat. Expression of these di-peptide repeats in cells interferes with multiple nuclear condensates including the nucleolus, splicing speckles and probably other charge-mediated condensates and cause toxicity[218,219]. Frameshifts in *HMGB1* cause the replacement of a highly acidic region with a basic sequence that leads to severe congenital defects[220]. Expression of the frameshift mutant proteins causes nucleolar dysfunction in cells. A survey of observed frameshift mutations in transcription factors and other proteins associated with pathogenic conditions, identified hundreds of frameshifts that create Arg-rich, basic C-terminal tail replacements[220] (Table 1), suggesting that this mechanism of condensate disruption by basic peptides could be a mechanism underlying other diseases.

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Changes in nuclear concentration of proteins

Abnormal overexpression of genes is associated with various diseases and in particular with cancers. Overexpression of proteins leading to increased protein concentration can result in the formation of new condensates that upregulate or downregulate the transcription of genes. A striking example of protein overexpression occurs through gene amplification; for example, the gene encoding transcription factor N-MYC is amplified in neuroblastoma, resulting in a 100-fold to 300-fold increase in protein concentration. The excess N-MYC forms nuclear condensates, driven by multivalent interactions between its intrinsically disordered transcription activation domain, which partition its partner MAX and other transcription machinery components such as MED1 and Ser5-phosphorylated Pol II[221]. These condensates boost the expression of oncogenes and repress the expression of tumour suppressor genes, thereby promoting cancer development[221] (Table 1). Similarly, overexpression of enhancer of zeste homologue 2 (EZH2), the catalytic subunit of the H3K27me3 methyltransferase PRC2, is associated with lung cancer[222]. In lung cancer cells, EZH2 undergoes N-myristoylation, which increases its hydrophobicity and promotes the formation of condensates, enabling the recruitment and methylation of H3K27 and STAT3. STAT3 methylation enhances STAT3 phosphorylation and activation, thereby further accelerating cancer progression[223,224] (Fig. 6f).

Aberrant cellular signals

Cellular signalling pathways are complex intracellular networks of molecules that can lead to diseases when dysregulated. Dysregulation may stem from transcriptional disturbances, which may further be attributed to abnormal phase separation, including the absence of functional condensates and the formation of abnormal ones. In normal physiology, the activation of the Hippo signalling pathway leads to phosphorylation of the transcription factors YAP and TAZ by LATS1 and LATS2, thus preventing their entry into the nucleus and subsequent gene activation. Conversely, when the pathway is deactivated, YAP and TAZ become active and translocate into the nucleus to regulate transcription[225]. Recent studies suggest that all of LATS2, YAP and TAZ can form condensates, potentially enhancing their function[226–231] (Table 1). In invasive breast cancer, FBXL16 overexpression inhibits the formation of LATS2 condensates, leading to YAP hyperactivation, which might promote cancer development[232]. Similarly, in liver cancer, glycogen condensates sequester the kinases MST1 (also known as STK4) and MST2 (also known as STK3), thereby hyperactivating YAP, which may accelerate tumorigenesis[233]. Another mechanism of YAP overactivation is the formation of YAP fusion proteins, which readily form condensates driven by multivalent interactions involving the newly fused IDRs. These condensates concentrate transcription cofactors and exclude repressive complexes, resulting in the formation of super-enhancers and long-range enhancer–promoter interactions, which activate genes that potentially drive cancer progression[234]. Similarly, in ADPKD, calcium-signalling-mediated cAMP hyperactivation causes serine/threonine-protein kinase SIK1 inactivation and hypo-phosphorylation of the transcription coactivator CRTC2. CRTC2 then accumulates in the nucleus and forms nuclear condensates that recruit P-TEFb, thereby enhancing transcription and promoting cyst growth[235] (Fig. 6g).

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Investigations of diverse diseases have converged on the dysregulation of condensate-mediated organization of the transcription machinery as a common cause of pathology. How the normal function of condensates is dysregulated in these different diseases highlights the principles that regulate them.

Conclusions and open questions

Using principles derived from polymer phase separation to investigate larger-scale organization of the transcription machinery has revealed a novel regulatory layer: dynamic selective compartmentalization through condensates. This regulatory layer operates at the 100–1,000 nm length scale by selectively organizing components of the transcription machinery either at specific genomic loci or away from the genome. Condensate formation largely depends on dynamic multivalent interactions seeded at specific genomic loci or off chromatin. That level of organization does not supersede the regulatory mechanisms operating at the 1–10 nm scale, but it rather works to promote or prevent them in the crowded environment of the nucleus. The formation and composition of condensates are regulated by tuning protein concentration, accessibility of regulatory elements, reversible multivalent PTMs, by ligand-induced conformational changes, and through other physiologically relevant parameters. These parameters are tuned during development to form condensates that activate robust yet switchable gene expression programmes. Condensates are responsive to environmental stimuli, providing a mechanism for rapid response and adaptation. The mechanisms that regulate transcriptional condensates can malfunction in various disorders and diseases.

Despite considerable advances in our understanding of how condensates regulate transcription, many unanswered questions and challenges remain. In this section, we highlight three major areas open for future investigation: (1) the rules of specificity in dynamic multivalent interactions, (2) the relationship between transcriptional condensates and enhancer–promoter specificity, and (3) strategies to therapeutically target transcriptional condensates.

The rules of specificity in dynamic multivalent interactions

Although some have argued that dynamic multivalent interactions are inherently nonspecific, recent experimental work discussed in this Review has shown that these interactions can be specific even when driven by disordered protein regions. A major open question is as follows: how many different ‘types’ or ‘flavours’ of specificity can be encoded by IDRs or other sources of dynamic multivalent interactions? Phrased another way, how many compositionally distinct condensates can coexist inside the nucleus or at different genomic loci? Proof-of-principle experiments have demonstrated that even subtle changes in sequence pattern and composition influence selective partitioning and that IDR sequences with identical composition, but modified patterning can largely differ in selective partitioning[84,101]. For example, experimentally rearranging the sequence without changing total composition of amino acids in the IDR of NELF, a negative regulator of transcription, caused it to gain partitioning into condensates composed of the $MED1^{IDR}$, leading to downregulation of transcription[84]. The reliance on multivalency, multiple points

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of contact, provides the potential for combinatorial complexity. Just considering IDRs, the diversity of potential combinatorial sequence features (for example, charge patterning, spacing of aromatic residues and multivalent motifs) even only among proteins involved in transcription is enormous. The extent to which this sequence diversity is used for condensate organization remains to be experimentally tested. We suggest that massively parallel reporter assays or other high-throughput methods coupled with functional readout (for example, expression or repression of transcription) will be needed to assess the degree of functional specificity encoded into IDRs and other sources of multivalent interactions. Of note, a theoretical work[236] has found that even when interactions between components are random, fluids containing multiple components segregate into many coexisting phases with distinct compositions.

Transcriptional condensates and enhancer–promoter specificity

What roles do condensates have in determining the specificity of regulatory elements? Does condensate compatibility or miscibility dictate enhancer–promoter interactions or other features of the 3D genome? Proof-of-principle experiments have demonstrated that genomic loci can be tethered together by the fusion of miscible condensates formed at different loci[237,238]. Genomic loci bound by the condensate-forming oncofusion NUP98–HOXA9 exhibit high contact frequency in HiC data, which is lost when the oncofusion is mutated to disrupt its condensate formation[204]. Strikingly, this condensate-mediated HiC contact is independent of CTCF and cohesin[204], proteins thought to be crucial for establishing contacts between genomic loci. Super-enhancers, which are sites of condensate formation[65], were found by multiple research groups to have heightened contact frequency with one another[239–241]. The distances between active enhancer and promoter pairs vary over a wide range with an average of ~300 nm (refs. 242,243), in striking agreement with the size of condensates formed in live cells at active genomic loci[81,82]. Together, these data suggest that the larger scale of organization of the transcription machinery could act as one of the underlying ‘tethers’ that guide enhancer–promoter specificity[244]. If that is the case, the specificity of condensate composition would direct which regulatory elements are a part of the same condensate and could be one of the mechanisms underlying selective cofactor dependencies of regulatory elements[245]. Methods that bridge high-resolution imaging of protein and DNA with genomics data are needed to investigate the relationship between condensates and enhancer–promoter specificity.

Strategies to therapeutically target transcriptional condensates

Given the multiple human diseases driven by dysregulation of condensate composition and the role of dynamic multivalent interactions in determining this composition, it is interesting to consider targeting dynamic multivalent interactions for therapeutics. Considerable attention has been given to the idea of ‘drugging condensates’ by using conventional small molecules[246,247]. With our current understanding that there are condensate-dependent interaction networks that utilize different types of interactions relative to one another, it might be possible to target these dynamic yet functional interactions. Standard methodologies in medicinal chemistry are firmly rooted in the structure–function paradigm, which is most relevant to organization at the 1–10 nm scale. What would a drug targeting the larger-scale network of interactions look like? One possibility would be to limit key nodes

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in the network to disrupt the formation of the network and/or prevent the partitioning of key proteins. For example, if enough dynamic multivalent interactions could be prevented, the formation of a condensate or the selective partitioning of a key factor could be inhibited. Initial progress has been achieved towards developing peptides or small molecules targeting signalling-driven[248] and androgen-receptor-driven[124,249] transcriptional condensates. Given the low affinities of any individual interaction, the affinity of the therapeutic may not need to be very high. Possibly, a drug designed to perturb specific networks could selectively target the activation of genes dependent on the specific condensates without affecting the expression of other genes. This approach might be most applicable to cases of oncogenic fusion proteins, repeat expansions, frameshift extensions or other instances in which disease cells have unique networks of interactions not found in healthy cells.

In summary, condensate-mediated organization of the transcriptional machinery provides novel regulatory opportunities not available to classic models of transcription. Regulation by condensate-mediated organization does not supersede the processes described by classic models, but it provides regulatory mechanisms for promoting or preventing these processes in the crowded environment of the nucleus. Although some of these condensate regulatory mechanisms have been described, we anticipate that continued investigations will reveal additional and surprising ways in which the cell takes advantage of phase separation principles in regulating transcription.

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Glossary

Critical thresholds

Specific concentrations or condition points at which a sudden transition occurs, such as phase separation.

DNA regulatory elements

DNA sequences located either proximal or distal to a gene that regulate its expression by interacting with transcription factors. Prominent examples are promoters and enhancers.

Domain–ligand pairs

Interacting modules between a protein domain and its corresponding ligand.

Histone locus body

A nuclear condensate associated with the active transcription of replication-dependent histone genes, which facilitates the coordination of histone mRNA transcription, processing and export during early S phase.

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Lower critical solution temperature behaviour

The phenomenon whereby a protein remains soluble in a solvent below a specific temperature, but above this temperature, it undergoes phase separation into condensates owing to the disruption of stabilizing interactions by increased thermal motion.

Mediator coactivator complex

A multiprotein complex that acts as a bridge between transcription factors and RNA polymerase II to regulate gene activation.

Multivalency

The property of a molecule or a complex that has multiple binding sites that can interact with multiple binding surfaces of another molecule or complex.

Patterned charge blocks

Sequences within proteins wherein charged amino acids are organized in clusters.

Phase separation

A physicochemical process through which molecules demix from the bulk environment to form dense, droplet-like structures.

Poly-glutamine

A sequence motif in proteins consisting of multiple Gln residues. Some poly-glutamine tracts enhance the formation of pathological condensates or aggregates by promoting strong hydrogen bonding and intermolecular interactions often associated with neurodegenerative diseases.

Re-entrant phase transition

A situation in a multi-component system in which increasing the concentration of a component initially favours the formation of condensates, but further increases lead to the disruption of the condensates.

Skotomorphogenesis

The developmental process in plants that occurs in darkness, characterized by elongated stems, closed cotyledons and lack of chlorophyll synthesis.

Transcription bursting

The phenomenon whereby transcription occurs in rapid, transient bursts rather than at a constant rate.

Transcriptional condensates

Biomolecular condensates that concentrate components of the transcription machinery and related factors.

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Box 1 |

Key steps in gene activation at the 1–10 nm length scale

Gene activation is regulated at many stages and requires the concerted action of dozens of proteins and nucleic acids at specific genomic loci. Here, we briefly summarize this process, which has been reviewed in detail elsewhere[1–6]. The accessibility of gene regulatory elements is highly regulated through the occlusion of transcription-factor-binding sites by individual nucleosomes and at a higher scale by chromatin structures that prevent activation of entire genomic regions. Some transcription factors, known as pioneer factors, can bind nucleosome-occluded DNA and recruit ATP-dependent chromatin remodellers, which make the regulatory element more accessible to binding by other transcription factors. These steps start the process of assembling transcription factors at specific genomic loci that further modify the chromatin, making it permissive for transcription and creating docking sites for other machinery. These docking sites include DNA, covalently modified histones, protein–protein interactions and RNAs. Multiple regulatory elements cooperatively assemble the general transcription factors that recruit the RNA polymerase II (Pol II) complex to promoters, thereby forming the pre-initiation complex (PIC). A sequence of enzymatic activities unwinds the DNA double helix, creating a transcription bubble that is engaged by Pol II. At many loci, as Pol II escapes PIC engagement into the gene, it is paused near the promoter by negative regulators of transcription. To release Pol II from this paused state, some of its subunits are phosphorylated, which promotes its binding by positive regulators and the formation of a transcript-elongation-competent enzyme. Many additional cofactors bind to and travel with Pol II to ensure processive elongation and proper co-transcriptional RNA processing. These processes have been reconstituted in vitro in the absence of any obvious higher-order clustering, demonstrating that they do not in principle require condensate formation. However, whenever these proteins are inspected at sufficient temporal and spatial resolution inside cells, they are found to form higher-order assemblies associated with activity. Although gene-specific studies have shown that high-order organization is required for transcription, whether all transcriptional events in cells require this high-order organization remains unclear. Single-cell genome-wide techniques that can relate condensate formation to transcription will be required to address this open question.

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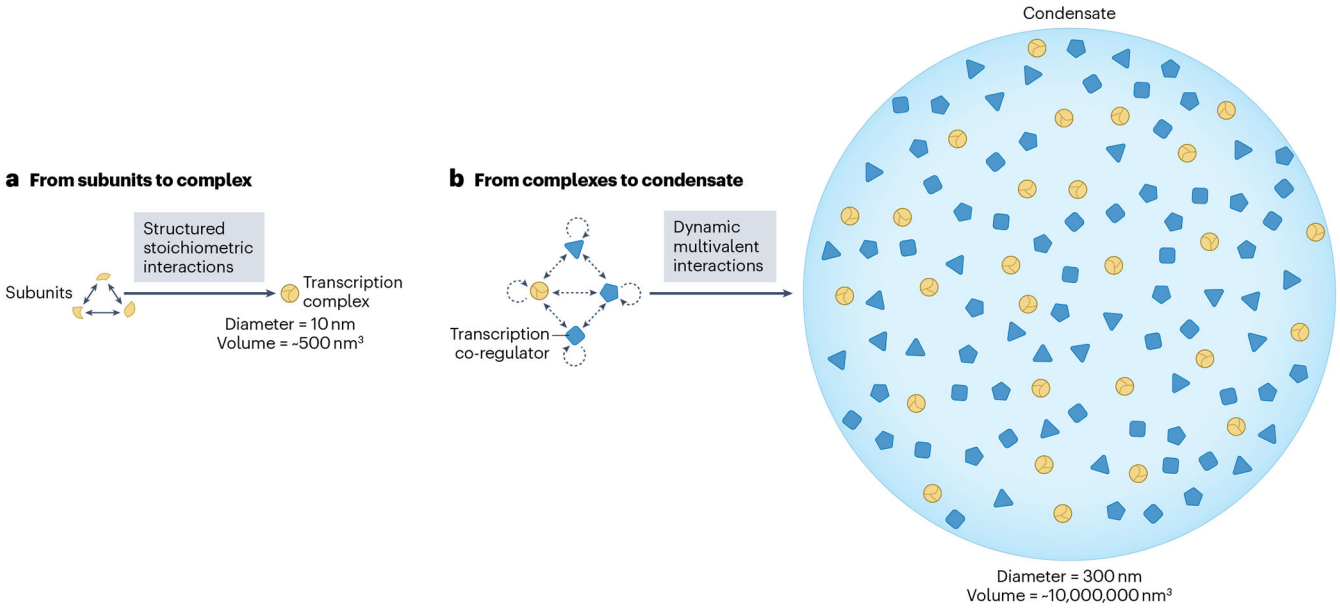

Fig. 1 |. Different length scales of biomolecular organization.

Key differences in the organization of biomolecules at the length scale of macromolecular complexes (1–10 nm) and biomolecular condensates (100–1,000 nm). **a**, Investigation of transcription regulation has been focused on understanding the organization of proteins and nucleic acids into macromolecular complexes at the 1–10 nm length scale. This organization level is driven predominantly by structured and stoichiometric interactions between complex subunits. A model transcription complex with a 10 nm diameter composed of three subunits that interact with each other through structured and stoichiometric interactions is shown as an example. **b**, Complexes can be further organized into condensates at the 100–1,000 nm length scale. This organization is driven predominantly by dynamic multivalent interactions. A model transcriptional condensate with a 300 nm diameter composed of the three-subunit complex and three other transcription complexes that interact with each other through dynamic multivalent interactions is shown. The purpose of these numbers and illustrations is to highlight the differences in scale of complex-mediated organization and condensate-mediated organization.

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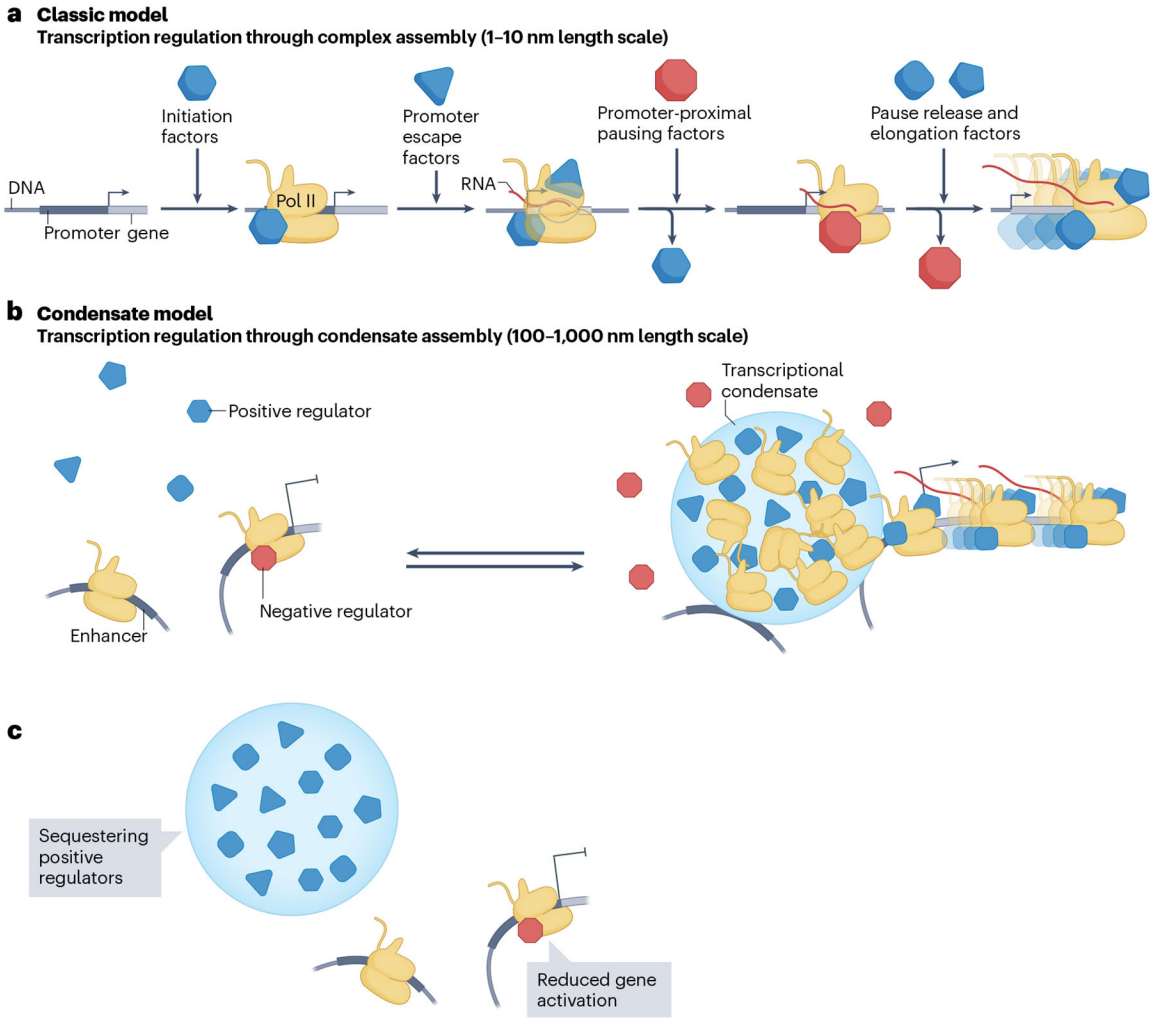

Fig. 2 |. Condensates provide organization at a larger scale.

a, A simplified schematic of the major steps of transcription by RNA polymerase II (Pol II) in the context of protein-complex formation. Classic models of transcription focus on the recruitment and allosteric regulation of Pol II by complexes, driven predominantly by structured and stoichiometric interactions. Distinct protein complexes form between Pol II and the protein factors required for each transcription stage with the final output being an elongation-competent Pol II. **b**, The condensate model of transcription focusses on how these transcription regulation complexes are organized at a larger scale (100–1,000 nm). The formation of such transcriptional condensates is driven predominantly by dynamic multivalent interactions. According to the condensate model, the transcription machinery is organized in cells to either facilitate or prevent gene activation. The simplified schematic shows how positive regulators, negative regulators, DNA and Pol II can organize into condensates to facilitate transcription. **c**, Organization of the transcription machinery by condensates can also result in reduced gene activation, as positive regulators of transcription can be sequestered into nuclear or cytoplasmic condensates, leading to reduced activation of genes that require these factors for their transcription.

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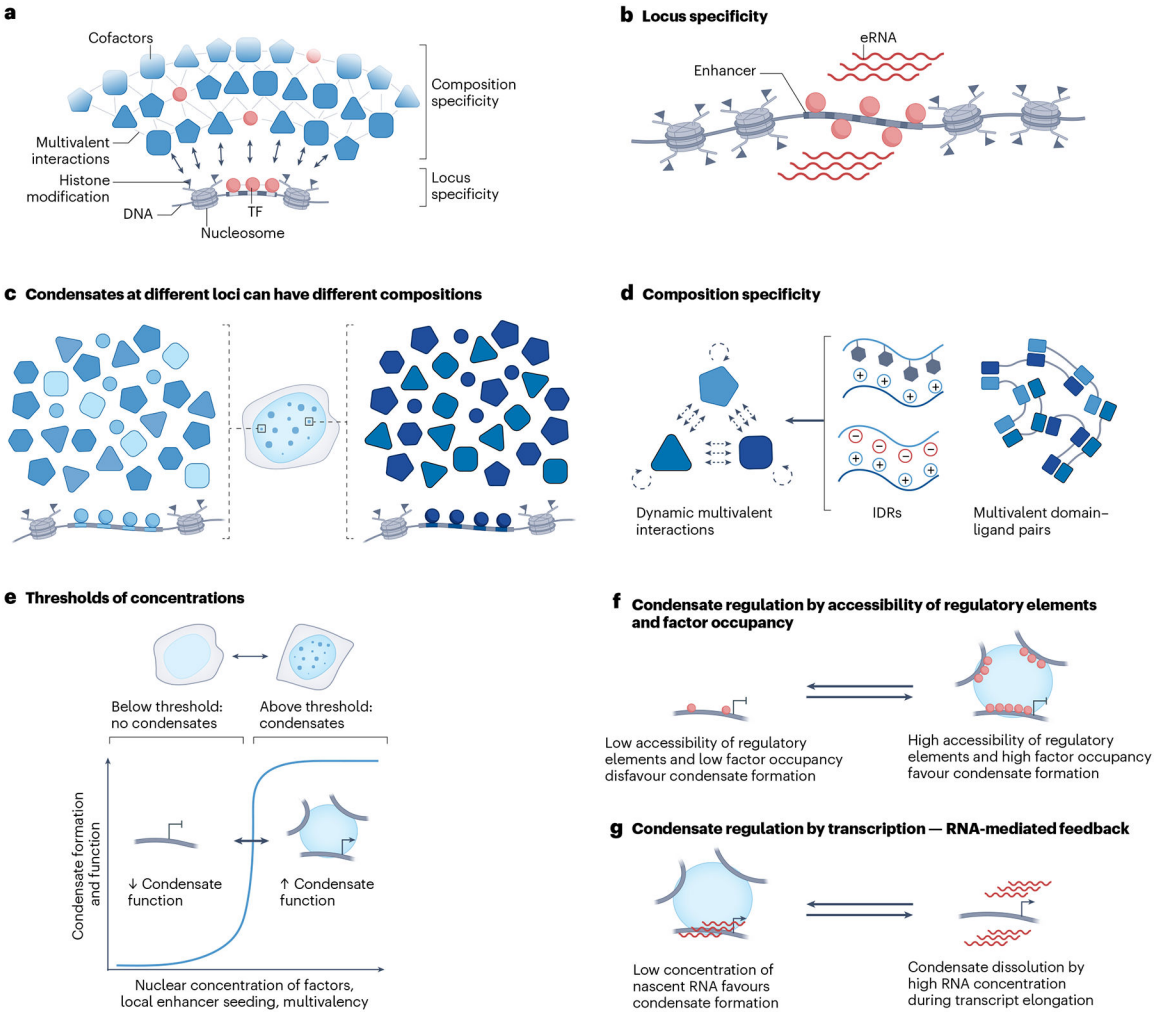

Fig. 3 |. Composition and formation of transcriptional condensates.

a, Transcriptional condensates are hyper-cooperative assemblies formed by both structured and dynamic multivalent interactions that form higher-order networks of biomolecules with unique compositions at specific genomic loci. Here, we highlight two distinct layers of condensate regulation that work together to determine the function of the condensate: composition specificity and locus specificity. This simplified schematic shows a genomic locus wherein transcription factors (TFs) seed or nucleate the formation of a network of interactions with a specific composition. Some of these interacting proteins are transcription cofactors that determine gene activation by RNA polymerase II (not shown). **b**, Transcriptional condensates form at specific loci and regulate gene expression. Chromatin-binding or DNA-binding proteins (pink circles) amplify the multivalency of specific genomic loci by simultaneously engaging with several biomolecules including DNA, RNA and proteins. Clustered regulatory elements such as enhancers and histone modifications are also multivalent features that determine and facilitate condensate formation. **c**, By using different combinations of locus and composition specificity, condensates with different compositions can form at different genomic loci, or off chromatin (not depicted), probably promoting diverse functions and mechanisms of gene activation control. **d**,

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The set of biomolecules recruited by TFs and local chromatin create a network of multivalent interactions that can selectively partition factors that do not have high-affinity interactions with TFs, cofactors or DNA. Certain multivalent sequence features in proteins such as positively charged and aromatic residues or patterned charge in intrinsically disordered regions (IDRs) can determine both selective partitioning into condensates and function. Multivalent domain–ligand pairs are another type of interaction that can confer composition specificity in transcriptional condensates. **e**, Transcriptional condensates offer unique regulatory mechanisms in gene expression. Condensates can function as molecular 'switches', as they form above certain thresholds of concentrations, availability of binding elements and multivalent interactions (*x*-axis). On the *y*-axis, we show how the increase of these features can lead to an increase in condensate formation and function (to activate or repress genes; gene activation shown). All these features (nuclear concentrations, availability of binding sites and multivalency) can be regulated to result in the assembly or disassembly of transcriptional condensates. **f**, Regulatory opportunities provided by accessibility of regulatory element and by factor occupancy. Insufficient factor concentration or regulatory-element accessibility will lead to low chromatin occupancy of factors, thereby disfavouring condensate formation. Increase in factor nuclear concentration or increase in regulatory-element accessibility will lead to chromatin occupancy of factors, which favours condensate formation. **g**, Regulatory opportunities provided by condensate-mediated re-entrant transitions in the context of transcribed RNA. Accumulation of nascent RNA favours condensate formation up to an optimal concentration. Increase in RNA concentration beyond the optimum, for example, during transcript elongation, causes imbalance in the network of multivalent interactions in the condensate and leads to condensate loss. eRNA, enhancer RNA.

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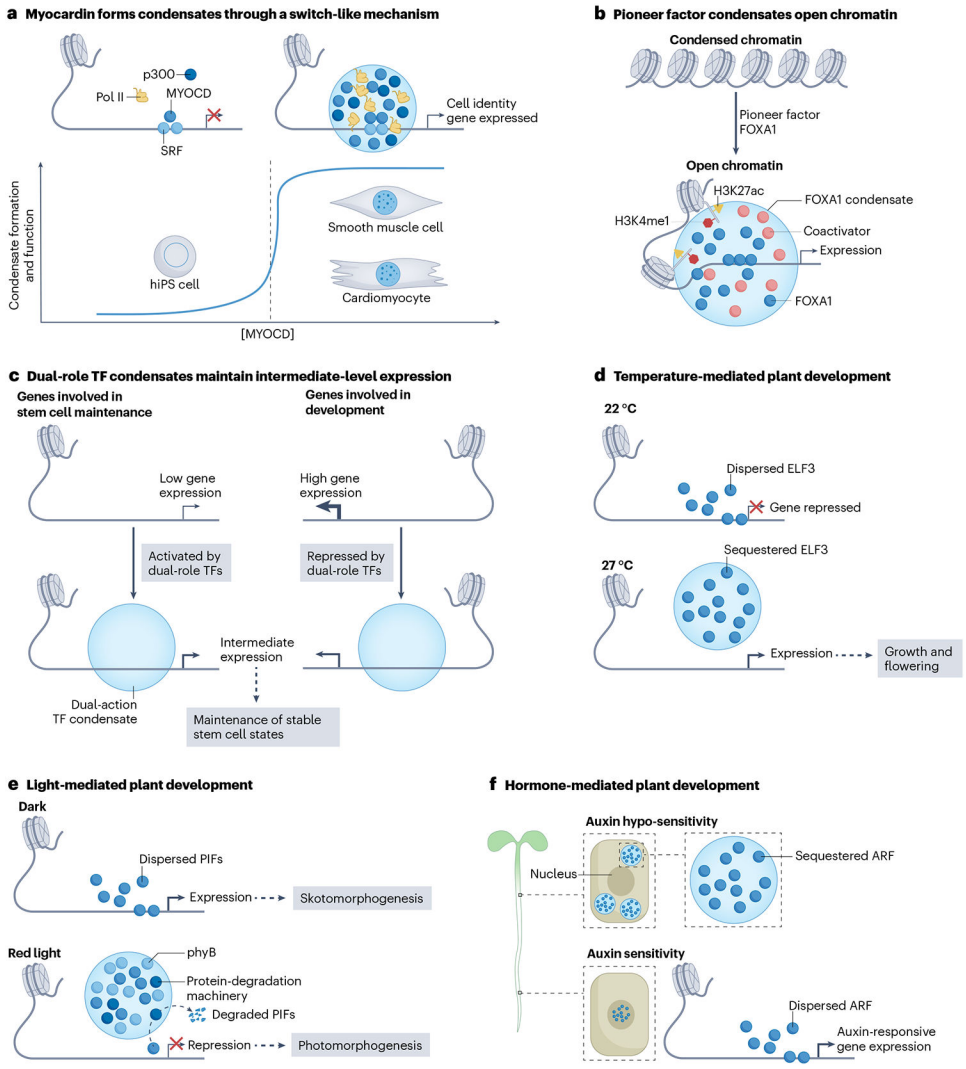

Fig. 4 |. Transcriptional condensates regulate various developmental processes.

a, In undifferentiated cells, such as human induced pluripotent stem cells (hiPS cells), the concentration of the transcription coactivator myocardin (MYOCD) is very low. In these cells, the transcription factor serum response factor (SRF), RNA polymerase II (Pol II) and the coactivator p300 are present, but they cannot initiate gene expression. Upon reaching the critical concentration, myocardin forms nuclear condensates that recruit Pol II and p300 to cell identity genes, leading to their activation and consequently driving cell differentiation. **b**, Pioneer factors such as forkhead box A1 (FOXA1) bind to condensed chromatin, open it and form condensates that attract transcription coactivators. These coactivators help to open the chromatin further by increasing the levels of acetylated histone H3 Lys27 (H3K27ac) and monomethylated histone H3 Lys4 (H3K4me1) and reducing H3K9me3 levels. This process ultimately upregulates the expression of target genes, particularly those involved in cell fate determination. **c**, Dual-role transcription factors (TFs) form nuclear condensates that decrease the expression of highly active genes, such as those involved in development, and increase the expression of lowly expressed genes, such as those involved in stem cell maintenance. This dual regulation maintains gene expression at intermediate

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levels, which ensures stability in stem cells. **d**, At 22 °C, EARLY FLOWERING 3 (ELF3) protein is dispersed within plant cells, attaching to DNA and repressing growth-related and flowering-related genes. At 27 °C, ELF3 forms condensates, thus sequestering itself away from DNA, lifting gene suppression and promoting plant growth and flowering. **e**, In dark, phytochrome-interacting factors (PIFs) activate genes for cell elongation and repress genes for chlorophyll biosynthesis and photosynthesis (not shown), leading to `skotomorphogenesis`. Exposure to red light triggers the phosphorylation and nuclear import of phytochrome B (phyB). In the nucleus, phyB forms photobodies that recruit PIFs and protein degradation machinery, leading to the degradation of PIFs and allowing plants to undergo light-mediated development (photomorphogenesis). **f**, In areas of mature roots with low auxin sensitivity, auxin response factors (ARFs) form cytoplasmic condensates that reduce their entry to the nucleus and gene activation. At the growing root tip, which exhibits high auxin sensitivity, ARFs remain dispersed in the nucleus, thereby promoting gene transcription for growth and development. The red X in panels **a**, **d** and **e** signifies that the gene remains unexpressed or is expressed at minimal levels.

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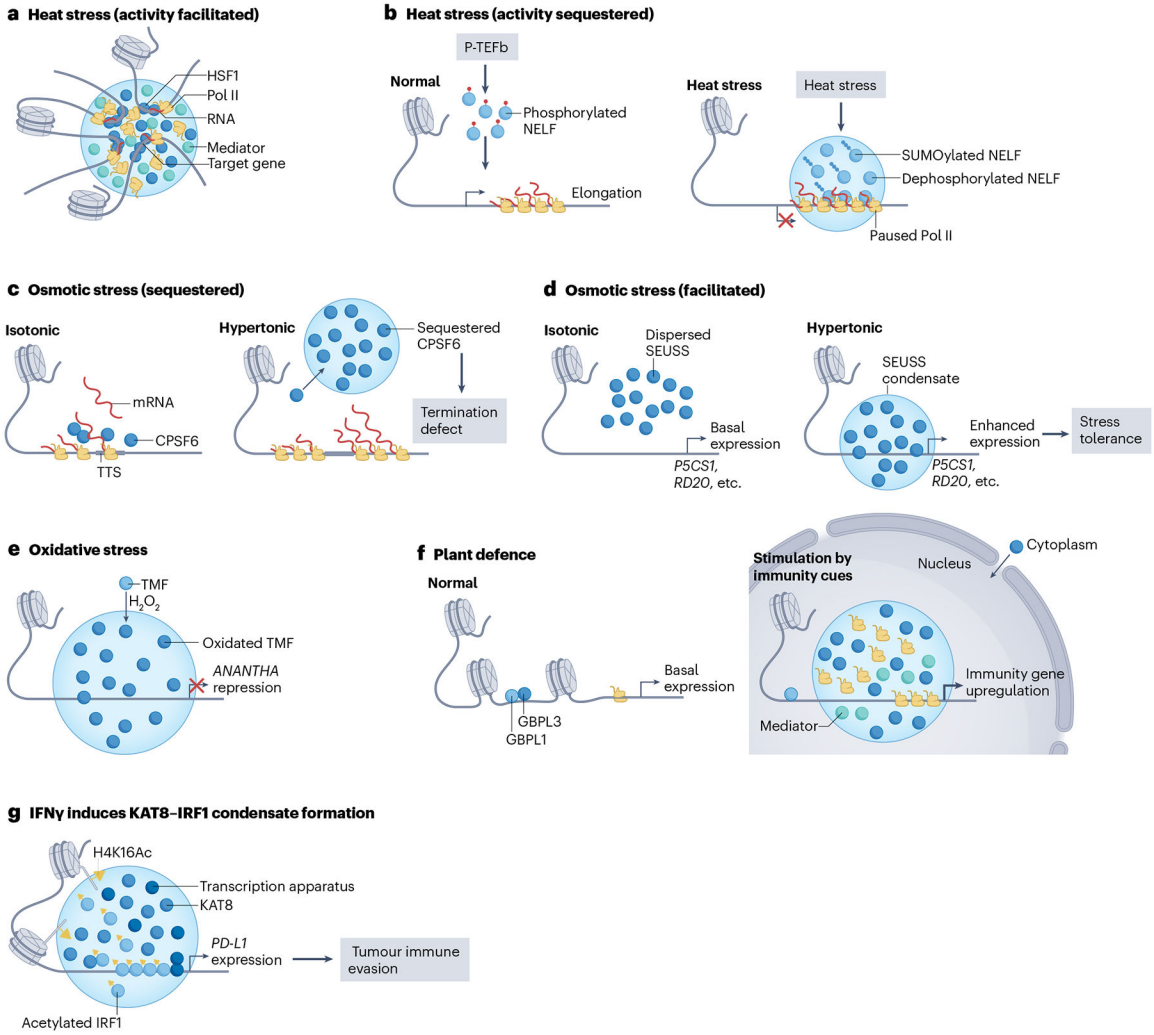

Fig. 5 |. Transcriptional condensates respond to environmental stimuli.

a, In response to heat shock, heat shock factor 1 (HSF1) forms dynamic condensates with Mediator and RNA polymerase II (Pol II). These condensates facilitate the coalescence of multiple HSF1 target genes across different chromatin domains and their expression. **b**, In normal conditions, negative elongation factor (NELF) is phosphorylated by positive transcription elongation factor b (P-TEFb) and releases Pol II for transcript elongation. Under heat stress, NELF is dephosphorylated and SUMOylated and forms condensates, thereby enforcing Pol II promoter-proximal pausing and transcriptional downregulation and ensuring cell viability. **c**, Cleavage and polyadenylation specificity factor 6 (CPSF6) binds sequences near the 3′-end of pre-mRNAs, thereby facilitating pre-mRNA cleavage and the addition of a poly(A) tail. Under hyperosmotic stress, CPSF6 forms condensates and detaches from actively transcribed genomic loci, thus globally impairing transcription termination. **d**, In normal conditions, the transcriptional regulator SEUSS is dispersed, and stress tolerance genes such as *P5CS1* and *RD20* maintain basal levels of expression. In response to hyperosmotic stress, SEUSS rapidly forms liquid-like nuclear condensates, enhancing the expression of these genes. **e**, The fast-growth-produced metabolite hydrogen peroxide (H_2O_2) triggers the formation of disulfide bonds (not shown)

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in the TERMINATING FLOWER (TMF) protein, which enable it to form condensates at the promoters of target genes such as *ANANTHA*, thus suppressing their expression and affecting floral development. **f**, In normal conditions, guanylate-binding protein-like GTPase 1 (GBPL1) binds to heterochromatin alongside GBPL3, thereby maintaining a basal level of transcription. Upon exposure to immune signals such as salicylic acid, pipecolic acid or pathogens, cytoplasmic GBPL3 becomes activated and forms nuclear condensates. These condensates facilitate the recruitment of Pol II and Mediator subunits, thereby upregulating immunity genes. **g**, In the tumour microenvironment, interferon-γ (IFNγ) triggers the formation of histone acetyltransferase KAT8–interferon regulatory factor 1 (IRF1) condensates, which boost the acetylation of histone H4 Lys16 (H4K16ac) and of IRF1, resulting in increased chromatin accessibility and high-affinity interactions with the *PDL1* promoter, which enhances *PDL1* transcription and facilitates tumour immune evasion. TTS, transcription termination site. The red X in panels **b** and **e** signifies that the gene remains unexpressed or is expressed at minimal levels.

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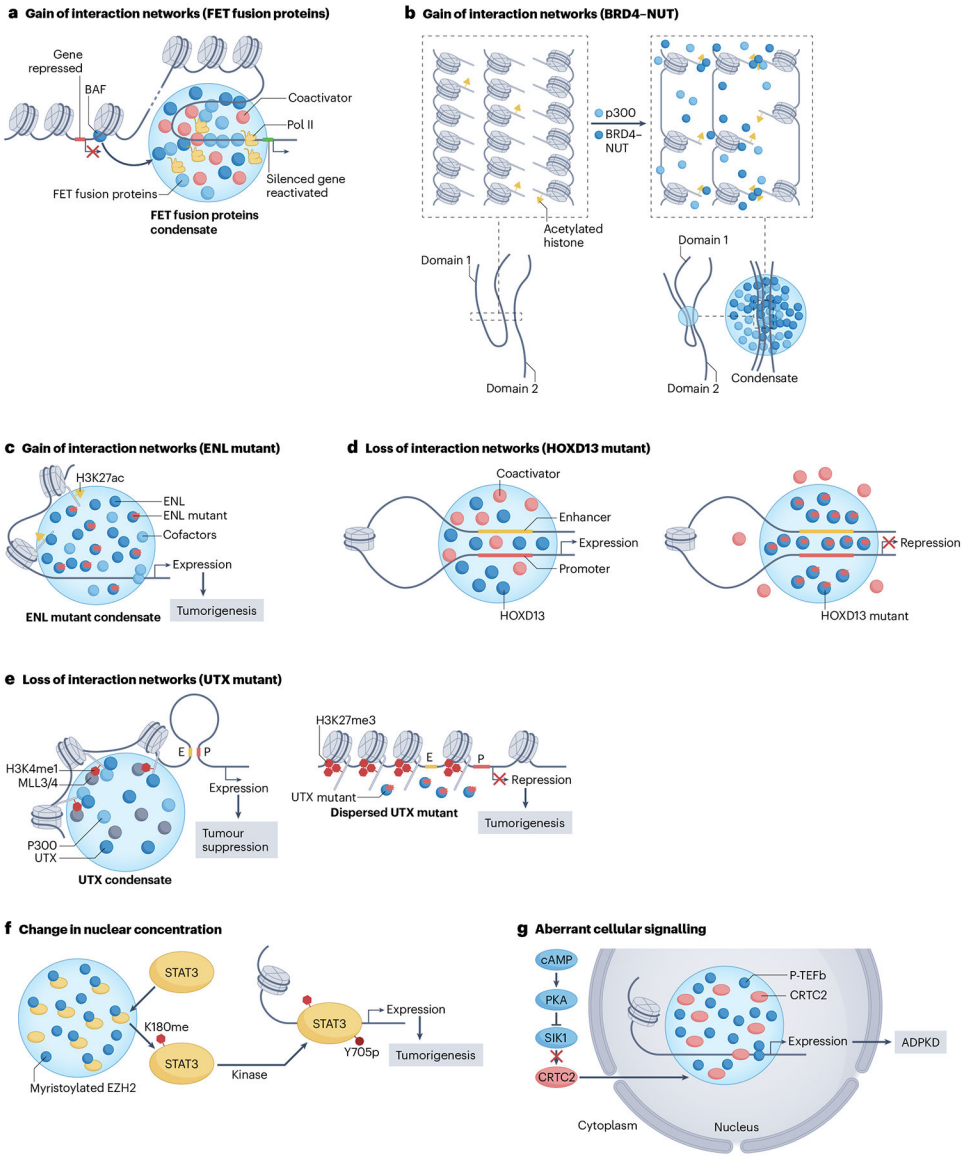

Fig. 6 |. Transcriptional condensates in disease.

a, FET (FUS, EWS, TAF15) fusion proteins can form dynamic condensates driven by the intrinsically disordered regions (IDRs) of the FET proteins. These condensates attract BAF chromatin remodeller complexes that open chromatin and recruit RNA polymerase II (Pol II) and transcription coactivators, thereby disrupting normal transcriptional processes, leading to reactivation of silenced genes and promoting tumorigenesis. **b**, BRD4–NUT fusion proteins bind to acetylated chromatin through the bromodomains of BRD4 (not shown) and form dynamic condensates driven by the IDR of NUT. The condensates recruit p300 to spread acetylation across extensive chromatin domains, ultimately leading to the formation of megadomains of open chromatin within the condensates. **c**, Mutation in the YEATS domain of ENL leads to the formation of abnormal condensates owing to enhanced self-interaction and interaction with transcription cofactors. These condensates localize at trimethylated histone H3 Lys27 (H3K27ac) chromatin and increase gene

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expression, particularly of the HOXA gene cluster, thereby contributing to cancer progression. **d**, Wild-type (left) and mutant homeobox D13 (HOXD13) (right) can both form nuclear condensates. However, mutant HOXD13 condensates fail to effectively attract transcription coactivators. This impairment leads to reduced expression of genes crucial for mesenchyme differentiation and digit formation. **e**, Wild-type ubiquitously transcribed X chromosome tetratricopeptide repeat protein (UTX) creates condensates that actively demethylate trimethylated histone H3 Lys27 (H3K27me3) and recruit the Lys methyltransferase MLL3 or MLL4 (MLL3/4) to boost H3K4 monomethylation (H3K4me1), thus facilitating transcription. These condensates collaborate with cofactors such as the histone acetyltransferase p300 in effectively regulating transcription of crucial tumour suppressor genes. By contrast, mutant UTX proteins fail to form these condensates, which leads to impaired gene regulation and contributes to tumorigenesis. **f**, In lung cancer cells, overexpressed enhancer of zeste homologue 2 (EZH2) undergoes N-myristoylation, which enhances its hydrophobicity, leading to the formation of nuclear condensates. These condensates recruit and methylate signal transducer and activator of transcription 3 (STAT3) at Lys180 (K180me), thereby facilitating its phosphorylation at Tyr705 (Y705p). This modification amplifies the transcriptional activity of STAT3, which promotes the expression of genes that accelerate cancer progression. **g**, In autosomal dominant polycystic kidney disease (ADPKD), increased cAMP signalling activates cAMP-dependent protein kinase A (PKA), which phosphorylates and inactivates serine/threonine-protein kinase SIK1. SIK1 inactivation leads to dephosphorylation of CREB-regulated transcription coactivator 2 (CRTC2), which then undergoes nuclear translocation and forms condensates that recruit positive transcription elongation factor b (P-TEFb), thereby enhancing the transcription of cyst growth genes. The red X in panels **a**, **d** and **e** signifies that the gene remains unexpressed or is expressed at minimal levels.

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Table 1 |

Key proteins involved in the formation of transcriptional condensates

Biomolecule	Functions through condensate formation	Refs.
A-kinase anchoring protein 95 (AKAP95)	AKAP95 condensates recruit SRSF2 and phosphorylated RNA polymerase II (Pol II), thereby influencing gene expression and promoting proliferation of breast cancer cells	250,251
Androgen receptor (AR)	Upon ligand activation and with the facilitation by MED1 (subunit of the Mediator coactivator complex), AR forms transcriptionally active condensates	124,183
Auxin response factors (ARFs)	In the mature upper root, ARFs form cytoplasmic condensates that sequester them, thereby reducing auxin-responsive gene transcription	85,165–167
Ataxin 2 (ATXN2/2L)	Ataxin-2 condensate formation oscillates with the circadian cycle, enriching translation initiation factors and the mRNAs of the circadian factors PER2 and CRY1, to upregulate their translation	252
At-rich interactive domain 1a/1b (ARID1A/B)	ARID1A/B, part of the mouse SWI/SNF chromatin remodeller complex, form condensates that recruit specific transcription cofactors	100
Fusion protein of bromodomain-containing protein 4 and nuclear protein in testis (BRD4–NUT)	BRD4–NUT binds to acetylated chromatin and forms condensates at different genomic domains, thereby recruiting p300 and altering chromatin structure, affecting gene expression and promoting tumour progression	205–208
β-Catenin	β-Catenin forms transcriptional condensates that aid gene activation through the compartmentalization of MED1 and Pol II; these condensates were shown to be dissolved by addition of short peptides derived from β-catenin cofactors such as FOXO4 and TCF	108,248
Chromodomain y like (CDYL)	CDYL downregulation maintains high histone crotonylation levels, thereby activating genes that support kidney cyst formation and causing autosomal dominant polycystic kidney disease (ADPKD), whereas, in healthy cells, CDYL condensates suppress these genes	253
CCAAT/enhancer-binding protein-α (C/EBPα)	C/EBPα forms condensates with transcription coactivators that promote long-range chromatin interactions, thereby facilitating *trans*-differentiation from B cells to macrophages	130
Cleavage and polyadenylation specificity factor 6 (CPSF6)	Under hyperosmotic conditions, CPSF6 forms condensates that sequester the pre-mRNA cleavage complex away from transcription-end sites, leading to readthrough transcription	173,174
CREB-regulated transcription coactivator 2 (CRTC2)	In ADPKD, CRTC2 dephosphorylation allows it to form nuclear condensates, which enhance the transcription of genes that promote cyst growth	235
Cryptochrome 2 (CRY2)	Under blue light, CRY2 forms condensates that recruit transcription factors such as TCP22, which boost the expression of the circadian transcription factor CCA1	161–164
CCCTC-binding factor (CTCF)	In human colorectal cancer cell lines, CTCF forms condensates that compartmentalize the transcription machinery and facilitate the formation of long-range chromatin loops and enhancer–promoter contacts In embryonic stem cells, CTCF forms condensates across different A compartments, thereby regulating the expression of pluripotency gene expression	135,254,255
Cyclin T1	DNA damage leads to poly(ADP-ribosylation) of cyclin T1, thereby preventing it from forming functional condensates and, thus, arresting transcript elongation	16,256
Damage-induced long non-coding RNAs (dilncRNAs)	DNA damage initiates synthesis of dilncRNAs, which form condensates with 53BP1 that aid DNA repair	257
Dishevelled 2 (DVL2)	Wnt signalling leads dishevelled-2 to form endogenous condensates, which activate β-catenin and promote target gene expression	258
Early B cell factor 1 (EBF1)	EBF1 forms nuclear condensates that recruit the chromatin remodeller complex BAF, which opens chromatin, and clusters EBF1-bound target genes, ensuring precise and robust gene programme establishment during cellular fate decisions	134

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Biomolecule	Functions through condensate formation	Refs.
Extrachromosomal circular DNA (ecDNA)	In glioblastoma neurospheres and prostate cancer cells, ecDNAs function as mobile super-enhancers, which attract Pol II and transcription cofactors to form condensates, thereby boosting genomic transcription and accelerating tumour cell evolution	230,259
E74-like factor 3 (ELF3)	In plants at 22 °C, ELF3 disperses in cells, binding to DNA and suppressing developmental genes; at 2 °C, ELF3 forms condensates, which reduce DNA binding, relieve gene suppression and promote growth	126,146,147
Eleven-nineteen-leukaemia (ENL)	Mutations in ENL promote its tendency to undergo condensate formation; ENL condensates recruit transcription cofactors and amplify gene expression, notably in the HOXA gene cluster, thereby contributing to cancer progression	95,115
Oestrogen receptor α (ERα)	ERα forms heterotypic condensates with MED1	92,105,184
Enhancer of zeste homologue 2 (EZH2)	In lung cancer cells, N-myristoylated EZH2 forms nuclear condensates, thereby recruiting and trimethylating histone H3 Lys27 (H3K27) and STAT3; STAT3 methylation enhances its phosphorylation, which increases its activity and promote cancer progression	223,224
Flowering control locus a (FCA)	FCA forms condensates that recruit RNA processing factors and, thereby, enhance RNA processing and ensure proper transcription termination	260
Forkhead box a1 (FOXA1)	FOXA1 forms biomolecular condensates to unpack condensed heterochromatin, initiating the expression of genes critical for cell fate determination	131–133
FRIGIDA	In cold conditions, FRIGIDA forms condensates, enhanced by COOLAIR, reducing *FLC* expression and preparing plants for flowering	151,152
FET (FUS, EWS, TAF15) protein fusions	FET protein fusions form condensates and recruit chromatin remodeller BAF complexes, Pol II and transcription coactivators, leading to aberrant transcription that promotes tumorigenesis	75,194–198
Guanylate-binding protein-like 3 (GBPL3)	Upon exposure to immune signals, GBPL3 activates and forms nuclear condensates that recruit Pol II and Mediator subunits to upregulate defence response genes	179
Glucocorticoid receptor (GR)	When bound by glucocorticoids, GR undergoes a conformational change and forms condensates at glucocorticoid response elements, thereby modulating transcription	125,180–182
High-mobility group box 1 (HMGB1)	HMGB1 proteins with a frameshift mutation form solid nuclear inclusions, which abnormally partition into the nucleolus and considerably reduce 28S ribosomal RNA levels and protein synthesis, impairing ribosome biogenesis and cell viability, and causing brachyphalangy, polydactyly and tibial aplasia syndrome, a rare complex malformation syndrome	220
Homeobox D13 (HOXD13)	A mutant HOXD13 transcription factor forms denser homotypic condensates with a diminished ability to recruit MED1, leading to decreased HOXD13 transcriptional activity and downregulation of genes crucial for mesenchyme differentiation and digit morphogenesis	212
hnRNP R-like protein (HRLP)	HRLP forms condensates that recruit the splicing factor SR45, which inhibits FLC pre-mRNA splicing, aids R-loop formation near FLC intron I, hinders Pol II recruitment to the FLC locus, suppresses *FLC* expression and promotes plant flowering	260
Heat shock factor 1 (HSF1)	Heat stress activates HSF1, forming nuclear condensates that recruit transcription machinery to facilitate the expression heat-shock factors, and boosting the transcription of satellite III long non-coding RNAs to suppress global gene expression	168,169
Interferon regulatory factors (IRFs)	The transcription factors IRF3 and IRF7 form condensates in innate antiviral immunity following their deacetylation by SIRT1	261
Jumonji domain-containing protein-3 (JMJD3)	TGFβ signalling in neural stem cells triggers JMJD3 to form condensates that induce chromatin reorganization and promote enhancer interactions, thus boosting gene activation	161
Lysine acetyltransferase 8 (KAT8)	KAT8 forms condensates with IRF1, which enhance H4K16 acetylation to open chromatin, thereby increasing IRF1 binding of the *PDL1* promoter and recruiting transcription factors to the gene, which aids tumour immune evasion	87
Krüppel-like factor 4 (KLF4)	During cell-type reprogramming, KLF4 forms condensates that help recruit transcription coactivators to pluripotency genes	137
Male-specific lethal 1 (MSL1)	Post-hepatectomy, MSL1 forms condensates in hepatocytes that compartmentalize STAT3, free histone 4 and acetyl-CoA, thereby increasing acetylation of H4K16 and STAT3K685ac to enhance STAT3 phosphorylation and activity and upregulate liver regeneration genes	88

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Biomolecule	Functions through condensate formation	Refs.
Myocardin	During differentiation of smooth muscle cells and cardiomyocytes, myocardin forms condensates that selectively recruit p300 and Pol II, culminating in targeted gene expression	83
Homeobox protein NANOG	During cell reprogramming, NANOG forms gel-like condensates that reshape the genome to enhance long-range interactions between different loci and super-enhancers, thereby activating genes essential for ground-state pluripotency	138
	During early zebrafish embryogenesis, NANOG forms condensates that sequentially attract Sox19b and Pol II, thus enhancing the expression of crucial developmental genes	145
Negative elongation factor (NELF)	Under heat stress, NELF quickly undergoes dephosphorylation and SUMOylation, forming condensates that extend Pol II promoter-proximal pausing, thereby downregulating transcription to preserve cell viability	89,172
N-MYC proto-oncogene (N-MYC)	In Kelly neuroblastoma cells, N-MYC forms nuclear condensates that recruit MAX and transcription machinery components, enhancing the transcription of oncogenes and inhibiting tumour suppressors, thus driving cancer progression	221
Nucleophosmin	Nucleophosmin forms pathological nuclear condensates that concentrate the transcription machinery and cofactors, thereby aberrantly activating HOXA/B and *MEIS1* genes and leading to leukaemic transformation	262
Nuclear factor erythroid 2-related factor 2 (NRF2)	In ADPKD, increased NRF2 phosphorylation causes NRF2 degradation and impairs NRF2 condensate formation and protection from oxidative stress	263
NUP98 fusions: NUP98–HOXA9, NUP98–KDM5A, NUP98–LNP1, NUP98–PRRX1	NUP98 fusion protein condensates promote chromatin looping, forming super-enhancer-like structures that recruit transcription coactivators or transcription factors, thereby boosting oncogene expression	200–204,264
Octamer-binding transcription factor 4 (OCT4)	During cell-type reprogramming, OCT4 condensates reorganize topologically associating domains, thereby redistributing super-enhancers and promoting the expression of pluripotency genes	136
ORESARA 1 (also known as NAC92)	Nitrogen deficiency induces the formation of ORESARA 1 and MED19a condensates, which boost cell-senescence-related gene transcription	265
p53	p53 forms condensates following DNA damage, which protect guarding against cancer; p53 mutations may yield diminished or less dynamic condensates and compromise protection	266–269
Transcription factor p65	In SK-N-AS cells exposed to TNF, p65 condensates with p300 and BRD4, synergistically promoting transcription	186
Peroxisome proliferator-activated receptor-γ coactivator 1α (PGC-1α)	PGC-1α condensates recruit phosphorylated Pol II and various transcription factors to H3K27ac-rich regions, thereby regulating glucose and fatty acid metabolism genes	270
Phytochrome b	Exposure to red or far-red light triggers phytochrome b to form photobodies in the nucleus, which attract protein degradation systems and phytochrome interacting factors (PIFs), leading to PIF degradation and promoting plant photomorphogenesis	154–160
Peroxisome proliferator-activated receptor-γ (PPARγ)	PPARγ, pairing with RXRα, forms transcriptionally active condensates that enhance gene activation	185
Polycomb repressive complex 1 (PRC1)	PRC1 subunits undergo phase separation and form condensates, driven by interactions between nucleosome arrays and different PRC1 subunits, and polymerization of PHC2	59,117,121,122
Reverse strand of ERBA-α (REV-ERBα)	REV-ERBα forms condensates that attract NCOR1 to repress circadian genes such as *BMAL1* and *NPAS2*, thereby regulating mammalian circadian rhythms	86,271
Ronin	Ronin forms condensates that assemble multiple promoters of housekeeping and metabolic genes, providing a mechanism to regulate their expression without the need for distal enhancer elements	272
SEUSS	Under osmotic stress, SEUSS forms condensates that regulate stress-responsive gene expression	176

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Biomolecule	Functions through condensate formation	Refs.
Splicing factor proline and glutamine-rich (SFPQ)	In human liver cancer, upregulated SFPQ forms nuclear condensates that sequester SMAD4, disrupting the SMAD activator complex and suppressing TGFβ-induced growth inhibition, thereby promoting carcinogenesis	273
SAGA complex associated factor 29 (SGF29)	In human mesenchymal progenitor cells and fibroblasts, SGF29 forms nuclear condensates at trimethylated H3K4 (H3K4me3)-rich sites, thereby recruiting transcription factors and coactivators to activate senescence genes	274
SS18 and SSX1 fusion oncoprotein (SS18–SSX1)	SS18–SSX condensates recruit the BAF complex, thereby enhancing gene expression and driving synovial sarcomagenesis	199
Shoot meristemless (STM)	STM forms nuclear condensates that attract the Mediator complex, thus amplifying transcriptional activity and enhancing the tolerance of plants to salt stress	275
Tata-box binding protein associated factor 14 (Taf14)	Yeast Taf14 forms condensates, which recruit transcription coactivators that act as a nuclear hub to dynamically regulate transcription and cellular pathways	276
TAF15	Human TAF15, with its extensive IDRs, forms condensates that recruit Pol II and, thus, regulate gene expression	78
TERMINATING FLOWER (TMF)	H_2O_2 induces Cys residues in TMF to form disulfide bonds, which enhance TMF condensation at the *ANANTHA* gene promoter, thereby suppressing its expression and affecting floral organ development	177,178
Ubiquitously transcribed X chromosome tetratricopeptide repeat protein (UTX)	UTX condensates demethylate H3K27me3 and recruit MLL3/4 and cofactors for tumour-suppressive transcription regulation; mutant UTX fails to form these condensates, thereby disrupting chromatin structure and tumour suppression	217
YAP protein fusions: C11ORF95-YAP, YAP-MAMLD1	YAP fusion proteins form nuclear condensates that concentrate transcription factors and coactivators, exclude PRC2 and activate genes that promote cancer progression	234
Yes-associated protein (YAP), transcription coactivator with pdz-binding motif (TAZ)	Unphosphorylated YAP and TAZ form nuclear condensates with transcription coactivators and phosphorylated Pol II, thereby activating transcription	226–231
Yin yang 1 (YY1)	YY1 condensates bridge enhancers and promoters, compartmentalize transcription coactivators and control long-range gene regulation	277
Zelda	In *Drosophila melanogaster* embryos, the formation of Zelda condensates facilitates the formation of Brd4 condensate, which then help initiate Pol II condensate formation and transcription	143,144